Jayanarundevaa Jayarajasekaran

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Profile

Biomedical engineering undergraduate with experience in laboratory testing, analytical instrumentation, and independent electrochemistry experimentation. Eager to contribute to multidisciplinary medical-device development through experimental problem-solving and materials-focused investigation.

Education

UNIVERSITY OF CALIFORNIA, IRVINE **Irvine, CA**

Bachelor of Engineering in Biomedical Engineering | Class of 2030 *September 2026 – Present*

- GPA: 4.0/4.0
- Relevant Coursework: Physics 7C, Math 2D

Experiences

CARBON ACTIVATED TESTING LABS **Compton, CA**

Lab Intern *June 2025 - August 2025*

- Supported activated-carbon quality testing by following ASTM standard procedures and established laboratory protocols.
- Gained hands-on exposure to ICP-MS and GC-MS analytical methods for chemical characterization.
- Operated UV-Vis and FT-IR instrumentation for qualitative and quantitative analysis.
- Conducted surface-area and porosity characterization using nitrogen adsorption isotherms and a Micromeritics TriStar II Plus.
- Followed laboratory safety procedures and handled chemical-testing equipment in a professional laboratory environment.

Skills

- **Analytical Instrumentation:** ICP-MS, GC-MS, UV-Vis, FT-IR
- **Material Characterization:** Activated-carbon testing, surface-area and porosity analysis, nitrogen adsorption
- **Laboratory Practices:** Chemical handling, laboratory safety, SDS/MSDS, equipment maintenance
- **Technical Skills:** C++, Java, SOLIDWORKS

Projects

SILVER-ON-COPPER ELECTROPLATING *January 2026 – March 2026*

Independent Material Science Project

- Conducted an independent electrochemistry experiment to deposit a silver coating onto a copper chain substrate.
- Investigated methods of suspension of chain to evaluate their effects on coating formation and uniformity, as well as different current and voltage combinations.
- Analyzed experimental observations to identify room for improvement in metal deposition through changes in methodology